Esercizio.

Suppose (X, τ) T2, compact topological space and let $A \in \tau \setminus \{\emptyset\}$. Then there exists $B \in \tau \setminus \{\emptyset\}$ such that $CL(B) \subseteq A$.

Dimostrazione.

Let $a \in A$. Consider $\{(U_x, V_x) : x \in A^c\} \subseteq \tau \times \tau$ such that $x \in U_x, a \in V_x, U_x \cap V_x = \emptyset$. Such a family exists because (X, τ) is T2. Now, since A^c is closed in a compact space we have that A^c is compact with the induced topology. Since $\{U_x : x \in A^c\}$ cover A^c we can extract a finite covering $\{U_i : i = 1, .., n\}$. So we have that $a \in \bigcap_{i=1,..,n} V_i \subseteq \bigcap_{i=1,..,n} U_i^c \subseteq A$. Since $\bigcap_{i=1,..,n} U_i^c$ is closed we conclude that $B := \bigcap_{i=1,..,n} V_i$ is the set we were looking for, since:

- it is non empty ($a \in B$)
- it is open
- $CL(B) \subseteq \bigcap_{i=1,..,n} U_i^c \subseteq A$

$\square$